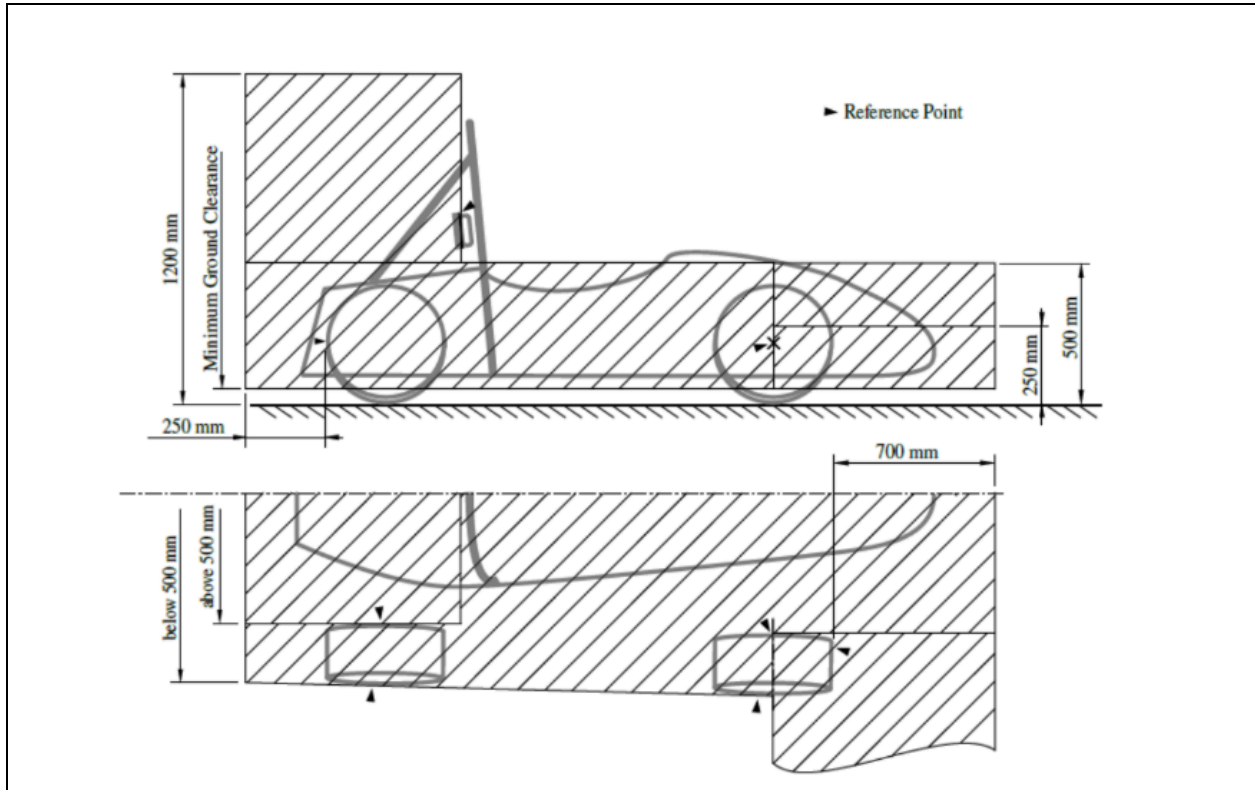


Guidelines and Rules on Aerodynamics and Bodyworks

Rulebook Restraints

- There should not be any random holes or openings apart from the top/upper space and those around front suspension or steering components, as minimal as possible. (T2.3.1)
- The bodywork in front of the cockpit should be smooth and convex (or flat), tightly fitted to have minimal spaces where they meet each other. (T2.3.2)
- Drainage holes on floor pan, at least 2 (T2.3.3)
- Bodywork in front of the front tyres that face upwards/forward at an angle greater than 45 degrees should have edges rounded to a minimum radius of 38 mm (1.5 inches). (T2.3.4)
- Aerodynamic Device: A specifically designed structure mounted on the vehicle to guide airflow around the vehicle, increasing downforce acting or lowering its drag. (T8.1.1)
- Height (T8.2.1):
 - Anything in front of the rearmost portion of the driver head restraint support (viz., the HANS device), without padding, in its rearmost position must be lower than 500 mm from the ground. Anything behind this position should be lower than 1.1 m from the ground.
 - Everything in front of the front axle and extending further outboard than the most inboard point of the front tyres must be lower than 250 mm from the ground.
- Width (T8.2.2):
 - All devices lower than 500 mm from the ground and rearward from front axle must not extend beyond the most outboard point of the front and rear wheels.
 - All devices higher than 500 mm must not extend beyond the most inboard point of the rear wheels.
- Length (T8.2.3):
 - All devices must not extend further rearward than 250 mm from the rearmost part of the rear wheels and must not extend further forward than 700 mm from the fronts of the front wheels.
- The restrictions must be fulfilled with the tyres pointing straight and the suspension setup, with or without the driver. (T8.2.4)
- Minimum radii of all bodywork and aerodynamic device edges that could contact a standing pedestrian without reaching the vehicle should be 3 mm for forward-facing ones and 1 mm for the rest. (T8.3.1)
- Any aerodynamic device must be able to withstand a force of 200 N distributed over a minimum surface area of 225 cm² and not deflect more than 10 mm in the load-carrying direction. (T8.4.1)
- Any device must be able to withstand 50 N of force, applied in any direction at any point and not deflect more than 25 mm. (T8.4.2)



Common Aerofoils

For front wing –

- Selig S1223
- Eppler E423
- Selig S1210 / S1223 RTL
- NACA 4412/415

For rear wing –

- For mainplane profiles:
 - Eppler E423 / Selig S1223
 - Liebeck LA2032 / Eppler E558
- For flap profiles:
 - Selig S1223 & Eppler E423
 - NACA 0012 / 0015 (Symmetric)

Approach to determine the most appropriate setup:

- Extract the coordinates from online catalogues, like:
 - airfoiltools.com
 - UIUC Applied Aerodynamics Database
- Run 2D planar simulations on the downloaded profiles, using:

- XFOIL
 - XFLR5
- Plot and understand the following curves:
 - Maximum downforce vs. angle of attack
 - Efficiency vs angle of attack
 - Stall behaviour (Downforce drop after peak; gradual is preferred)
- After shortlisting preferred candidates, perform full analysis on 3D CFD platforms (like Fluent):
 - **For Front Wings:** 2D XFOIL data overestimates performance because it doesn't account for ground proximity. Once you narrow down 2 or 3 top candidate profiles in 2D, bring them into 3D CFD (e.g., ANSYS Fluent, OpenFOAM, or Star-CCM+) placed at your actual ride height (25–50 mm) to verify ground-effect pressure distributions.
 - **For Rear Wings:** Test candidates in 3D CFD with your targeted multi-element gaps and overlaps. Multi-element aerofoils allow high energy flow from the pressure side to re-energise the boundary layer on the suction side, preventing low-Re separation.
- For multi-element rear wing setups, XFLR or XFOIL are not very helpful; it is better to rather rely on 2D RANS solver on Ansys Fluent or OpenFOAM, which accommodates other multi-element effects.
- Driving velocity also brings in certain changes and variations to the parameters, so it is beneficial to follow this sequence:
 - Map operational velocity ranges – cornering, braking and maximum (straightway)
 - Calculate variable Re spectrum
 - Convert coefficients to actual forces
 - Evaluate pitch and velocity sensitivity
 - Reject any aerofoil profile whose stall angle drops abruptly when moving from high Re (28 m/s) to low Re (10 m/s). You want a **soft stall profile** that maintains a gentle downforce delivery across all operating speeds.